

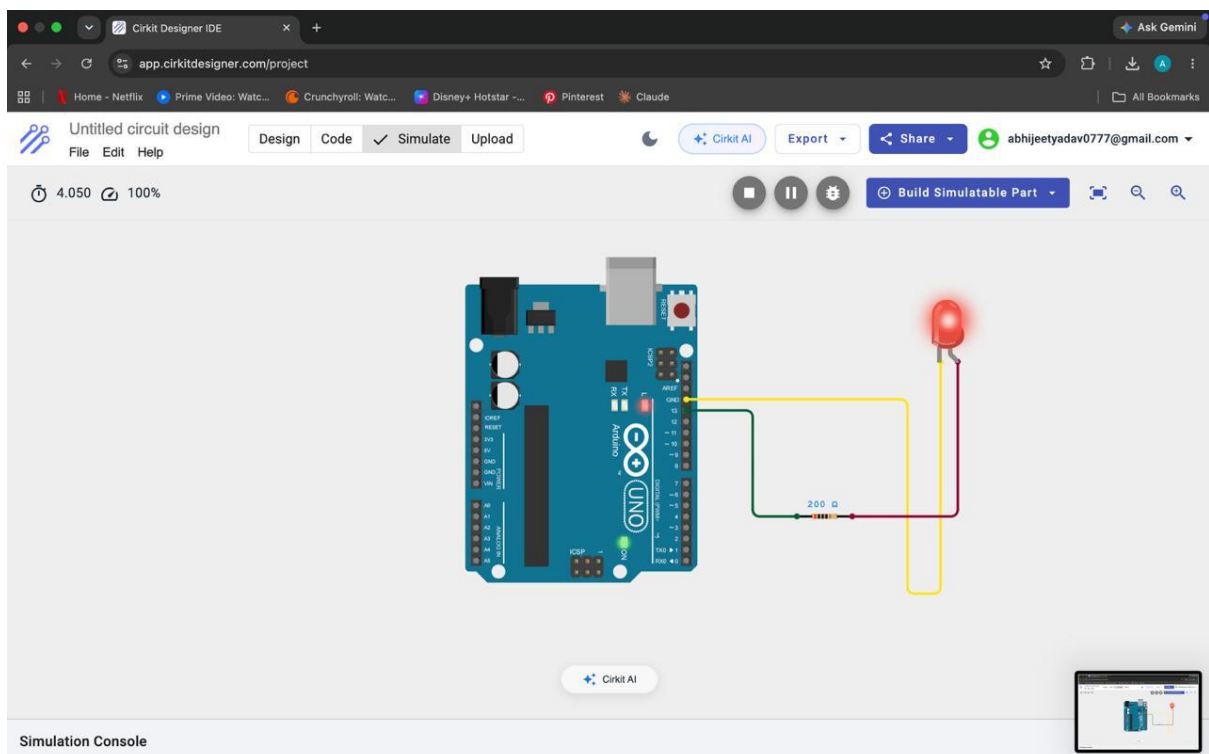
1.Blink an LED

#source code

```
void setup() {  
  pinMode(13, OUTPUT);  
}
```

```
void loop() {  
  digitalWrite(13, HIGH);  
  delay(1000);  
  digitalWrite(13, LOW);  
  delay(1000);  
}
```

Output



2.Control using push button

#source code

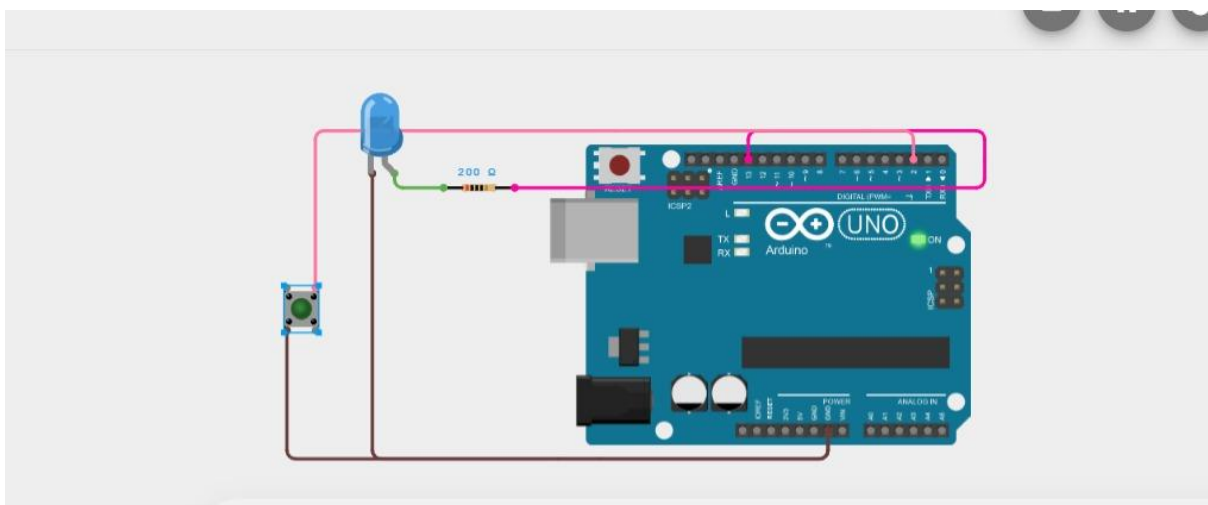
```
const int ledPin = 13;

const int buttonPin = 2;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(buttonPin, INPUT_PULLUP);
}

void loop() {
  if (digitalRead(buttonPin) == LOW) {
    digitalWrite(ledPin, HIGH); // LED ON
  }
  else {
    digitalWrite(ledPin, LOW); // LED OFF
  }
}
```

#output

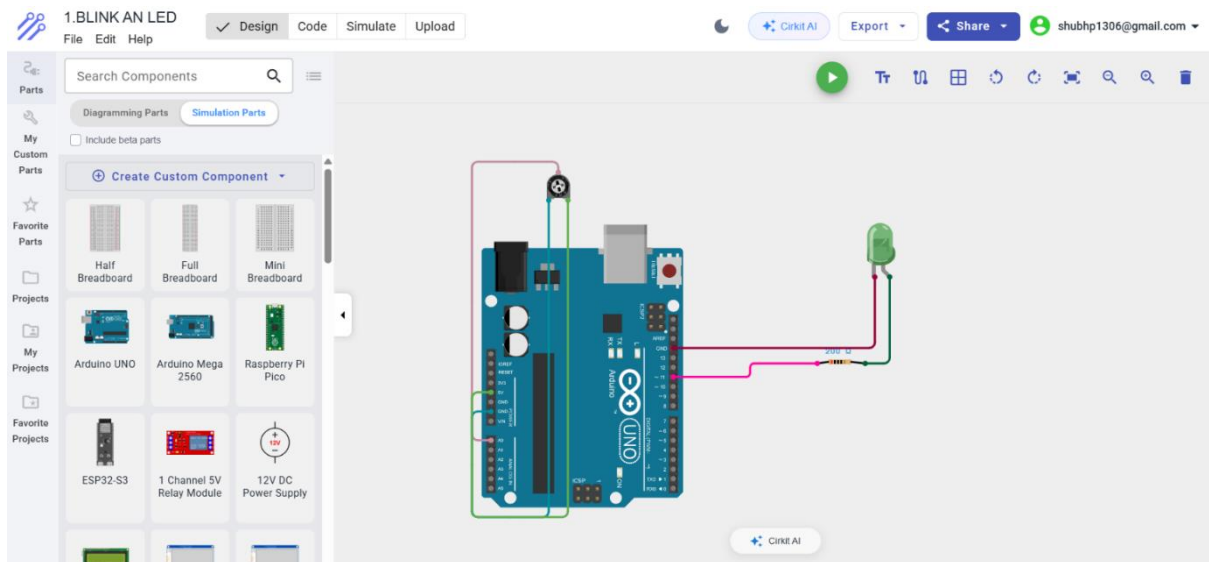


3.Increase or decrease the LED illumination

#source code

```
const int potPin = A0;  
const int ledPin = 11;  
int potValue = 0;  
int brightness = 0;  
void setup() {  
  pinMode(potPin, INPUT);  
  pinMode(ledPin, OUTPUT);  
}  
void loop() {  
  potValue = analogRead(potPin);  
  brightness = map(potValue, 0, 1023, 0, 255);  
  analogWrite(ledPin, brightness);  
  delay(10);  
}
```

#output



4. Display A OBJECT using IR SENSOR AND DISPLAY IT ON SERIAL MONITOR .

#source code

```
const int irPin = 2;
const int ledPin = 13;

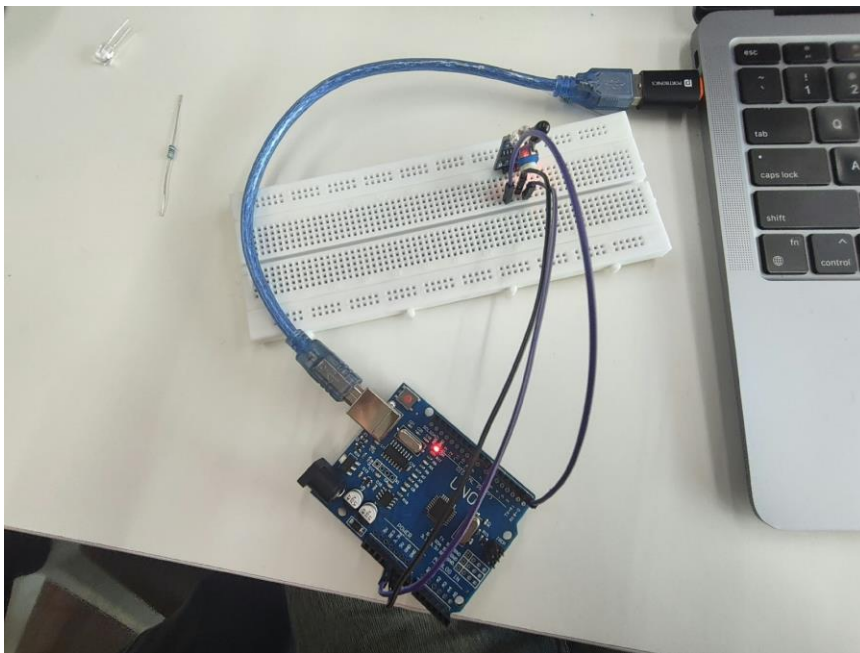
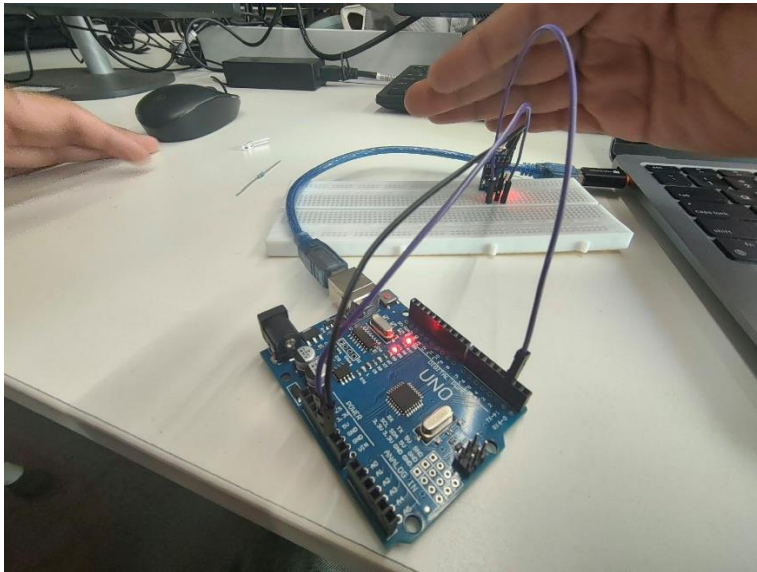
void setup() {
  pinMode(irPin, INPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {

  int sensorState = digitalRead(irPin);
  if (sensorState == LOW) {
    digitalWrite(ledPin, HIGH);
    Serial.println("Object detected!");
  }
  else {
    digitalWrite(ledPin, LOW);
    Serial.println("No object");
  }

  delay(200);
}
```

#output



5. To measure the distance of an object and display the distance on the Serial Monitor.

#source code

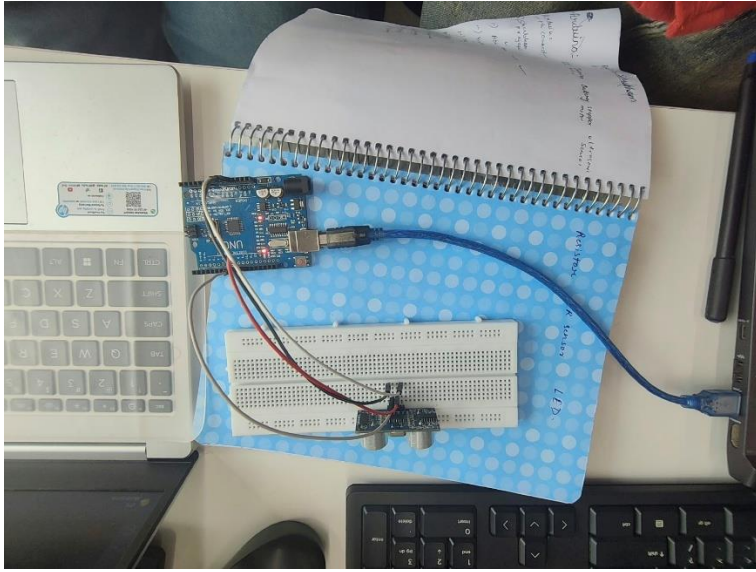
```
const int trigPin = 9;
const int echoPin = 10;

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  // Clear the trigPin
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH);
  float distance = duration * 0.034 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(500);
}
```

#output



6. Connect the DHT11 sensor data pin to digital pin 2, and connect the OLED display via I2C to the Arduino."

#SOURCE CODE

```
#include <Wire.h>
```

```
#include <Adafruit_GFX.h>
```

```
#include <Adafruit_SSD1306.h>
```

```
#include <DHT.h>
```

```
#define SCREEN_WIDTH 128
```

```
#define SCREEN_HEIGHT 64
```

```
#define OLED_RESET -1
```

```
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,  
OLED_RESET);
```

```
#define DHTPIN 2
```

```
#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  dht.begin();
```

```
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
```

```
    Serial.println(F("OLED allocation failed"));
```

```
    for(;;);
```

```
  }
```

```
  display.clearDisplay();
```

```
  display.setTextSize(1);
```

```
  display.setTextColor(SSD1306_WHITE);
```

```
}
```

```
void loop() {
```

```
  delay(2000);
```

```
  float h = dht.readHumidity();
```

```

float t = dht.readTemperature();
if (isnan(h) || isnan(t)) {
    Serial.println(F("Failed to read from DHT sensor!"));
    return;
}
display.clearDisplay();
display.setCursor(0, 0);
display.println(F("Live Readings:"));
display.setCursor(0, 20);
display.print(F("Temp: "));
display.print(t);
display.println(F(" C"));
display.setCursor(0, 40);
display.print(F("Humidity: "));
display.print(h);
display.println(F(" %"));
display.display();
}

```

#output

